

TURKU AP, Finland

WMO: 029720

Lat: 60.516N

Lon: 22.279E

Elev: 192

StdP: 14.59

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-5.8	0.0	-9.7	3.3	-5.3	-3.8	4.5	0.6	21.6	28.0	19.4	29.3	4.4	70	0.621

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	80.8	64.4	77.2	63.1	73.8	61.4	67.4	76.0	65.4	73.3	63.5	71.1	8.9	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.5	91.6	70.4	62.5	85.2	68.6	60.6	79.4	66.9	31.9	75.9	30.3	73.0	28.9	71.1	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.7	15.0	DB	-10.8	84.4	6.7	3.6	-15.6	87.0	-19.6	89.1	-23.3	91.1	-28.2	93.7	(4)
(5)			WB	-10.8	69.4	6.8	2.3	-15.7	71.1	-19.6	72.5	-23.4	73.8	-28.4	75.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.1	23.9	23.1	29.2	39.6	49.3	57.2	63.1	60.8	52.4	42.3	34.5	28.7
	DBStd	15.78	10.56	10.88	7.76	6.17	7.11	5.82	5.35	5.34	5.94	7.13	7.98	9.91
	HDD50	4050	810	752	646	318	98	6	0	1	43	251	465	660
	HDD65	8427	1275	1172	1111	763	488	243	101	152	378	704	915	1125
	CDD50	1173	0	0	0	5	77	222	407	335	115	12	0	0
	CDD65	75	0	0	0	0	2	9	42	22	0	0	0	0
	CDH74	719	0	0	0	0	37	112	402	167	1	0	0	0
	CDH80	126	0	0	0	0	3	13	89	21	0	0	0	0
Wind	WSAvg	7.1	7.4	7.1	7.1	7.1	7.2	7.3	6.8	6.5	6.7	7.1	7.5	7.7
Precipitation	PrecAvg	28.5	2.4	1.7	1.7	1.5	1.4	2.0	2.9	3.2	2.7	2.8	3.1	2.8
	PrecMax	38.0	4.3	4.2	4.1	4.0	3.5	4.1	5.7	7.5	6.4	6.8	6.4	7.2
	PrecMin	20.9	0.4	0.1	0.2	0.1	0.2	0.3	0.0	0.0	0.5	0.8	0.5	0.3
	PrecStd	3.8	1.1	1.0	0.9	0.9	0.9	1.1	1.6	1.7	1.4	1.3	1.5	1.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.4	41.2	51.6	66.0	78.8	82.0	86.4	82.7	71.6	59.2	50.1	46.6
		MCWB	40.6	37.9	42.2	51.9	61.4	63.4	67.2	64.9	60.1	54.1	49.0	45.4
	2%	DB	39.4	39.0	46.0	60.5	73.2	77.3	82.5	78.6	67.9	55.8	48.3	44.5
		MCWB	38.4	37.2	39.7	48.4	57.5	61.2	65.6	64.2	59.2	52.8	47.2	43.7
	5%	DB	37.6	37.2	41.4	55.8	68.3	73.3	78.6	75.1	64.7	53.9	46.6	42.5
		MCWB	36.7	36.1	37.0	45.7	55.2	59.5	64.3	62.9	57.6	51.5	45.5	41.4
	10%	DB	35.8	35.5	38.9	51.8	63.4	69.6	75.1	71.7	62.3	52.1	44.8	40.6
		MCWB	35.0	34.7	35.5	43.6	52.4	57.6	63.0	61.4	56.8	49.9	43.8	39.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.2	38.9	43.6	52.8	63.9	67.0	70.4	69.2	62.3	55.9	49.4	45.6
		MCDB	41.3	39.8	48.8	62.8	72.8	77.2	79.9	76.9	67.7	57.6	49.6	45.8
	2%	WB	38.5	37.3	40.0	49.8	60.0	63.6	68.3	66.8	60.6	53.9	47.3	43.4
		MCDB	39.0	38.0	43.9	58.5	69.8	73.0	78.1	74.0	65.7	55.7	47.8	43.9
	5%	WB	36.7	35.8	37.7	46.9	56.6	61.1	66.2	64.9	59.2	52.3	45.6	41.1
		MCDB	37.2	36.6	40.5	54.0	66.4	69.6	74.5	71.6	63.3	54.0	46.2	41.7
	10%	WB	34.9	34.4	35.9	44.3	53.3	59.0	64.3	63.1	57.6	50.7	43.8	39.1
		MCDB	35.5	35.1	37.9	50.3	62.4	67.1	72.4	69.6	61.3	52.3	44.5	39.7
Mean Daily Temperature Range	5% DB	MDBR	9.6	10.6	13.4	16.6	18.5	17.2	17.1	17.1	15.0	11.0	7.9	8.6
		MCDBR	7.9	9.0	15.9	24.4	26.0	24.2	23.3	22.0	17.9	11.4	7.7	8.0
		MCWBR	7.7	7.8	11.0	14.3	13.3	11.5	9.9	10.0	10.5	8.7	7.6	7.7
	5% WB	MCDBR	7.4	8.2	13.6	21.7	22.7	20.2	19.7	17.3	15.1	9.2	7.6	7.4
		MCWBR	7.3	7.5	10.1	13.6	12.5	10.9	9.8	9.2	9.7	7.7	7.7	7.4
Clear-Sky Solar Irradiance	taub	0.273	0.311	0.331	0.370	0.366	0.358	0.386	0.376	0.347	0.330	0.298	0.382	
	taud	2.148	2.236	2.316	2.327	2.403	2.450	2.394	2.417	2.485	2.400	2.210	2.614	
	Ebn at Noon	149	213	251	262	273	277	265	258	246	207	138	103	
	Edh at Noon	13	22	28	33	33	32	33	30	24	19	12	9	
All-Sky Solar Radiation	RadAvg	76	261	739	1233	1679	1779	1669	1309	803	348	96	45	
	RadStd	13	51	89	121	160	140	165	134	99	57	18	5	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (59 neighbors)		+0.74	N/A	N/A	N/A	N/A	-239	N/A	N/A	N/A

Nomenclature: See separate page